

LINEAR AND ROBUST CONTROL SYSTEMS

ELEC 9731

9B - Polynomial Matrices and Minimal State
Space Realizations

© Prof. Victor Solo

School of Electrical Engineering and
Telecommunications

UNSW

Sydney

Slide 1

Topics

- 1** Smith Form
- 2** Smith-McMillan Form
- 3** Kalman SS form

References

Kailath ch 6

Smith Form

- Given a $p \times m$ polynomial matrix $N(s)$ of generic rank r . It can be reduced to a pseudo-diagonal form by elementary row and column operations. i.e. there are unimodular matrices $U(s), V(s)$ such that

$$U_{p \times p}(s)N(s)_{p \times m}V_{m \times m}(s) = \Lambda_{p \times m}(s)$$

$$\Lambda(s) = \begin{matrix} & \begin{bmatrix} \lambda_1(s) & 0 & \cdots & 0 \\ 0 & \cdots & \cdots & \cdots \\ \cdots & \cdots & \lambda_r(s) & 0 \\ 0 & \cdots & 0 & 0 \end{bmatrix} & \begin{matrix} r \\ m-r \end{matrix} \end{matrix}$$

r
 $p-r$

$\lambda_i(s)$ are polynomials satisfying a division property

$$\lambda_i(s) \mid \lambda_{i+1}(s), i = 1, \dots, r-1$$

Slide 3

SMITH FORM Continued

In fact if $\Delta_i(s) = \text{gcd of all } i \times i \text{ minors of } N(s)$
then

$$\lambda_i(s) = \frac{\Delta_i(s)}{\Delta_{i-1}(s)}, \Delta_0(s) = 1$$

$\Lambda(s)$ is unique but $U(s), V(s)$ are not.

Note. we can also write then

$$N(s) = \overline{U}(s)\Lambda(s)\overline{V}(s)$$

where $\overline{U}(s), \overline{V}(s)$ are unimodular. This is often referred to as the Smith form.

$\lambda_i(s)$ are called the invariant polynomials of $N(s)$.

SMITH FORM Continued II

Proof

- (i) Bring the element of least degree in $N(s)$ to the 1,1 position by row,col interchange.
- (ii) By elementary column operations zero all other entries in the first column.
- (iii) By elementary column operations zero all other entries in first row. In so doing may get nonzero entries in first column. So repeat above process.
- (iv) If this 1,1 element does not divide all others use division and row, column interchange to bring lowest degree element to the 1,1 position. Repeat above procedure.
- (v) This leaves the form
$$\left[\begin{array}{c|c} \lambda_1(s) & 0 \\ \hline 0 & N_1(s) \end{array} \right]$$
and $\lambda_1(s)$ divides all elements in $N_1(s)$
- (vi) Repeat the process on $N_1(s)$
- (vii) Iterate to get the result.

Example 9.3

$$P(s) = \begin{bmatrix} \frac{1}{(s-1)^2} & \frac{1}{(s-1)(s+3)} \\ \frac{-6}{(s-1)(s+3)^2} & \frac{s-2}{(s+3)^2} \\ \frac{s}{(s+3)(s-1)^2} & \frac{1}{(s-1)(s+3)} \end{bmatrix}$$

$$\Rightarrow d(s) = (s-1)^2(s+3)^2$$

$$\Rightarrow N(s) = \begin{bmatrix} (s+3)^2 & (s-1)(s+3) \\ -6(s-1) & (s-2)(s-1)^2 \\ s(s+3) & (s-1)(s+3) \end{bmatrix}$$

Example 9.3 Continued

Form gcds

$$\Delta_1(s) = \gcd (\text{all elements}) = 1$$

$$\Delta_2(s) = \gcd (\text{all } 2 \times 2 \text{ minors})$$

$$= \gcd \left\{ \begin{vmatrix} (s+3)^2 & (s-1)(s+3) \\ -6(s-1) & (s-2)(s-1)^2 \end{vmatrix}, \right. \\ \left. \begin{vmatrix} (s+3)^2 & (s-1)(s+3) \\ s(s+3) & (s-1)(s+3) \end{vmatrix}, \right. \\ \left. \begin{vmatrix} -6(s-1) & (s-2)(s-1)^2 \\ s(s+3) & (s-1)(s+3) \end{vmatrix} \right\} \\ = \gcd \left\{ \begin{array}{l} (s+3)(s-1)^2((s+3))(s-2) + 6, \\ (s+3)^2(s-1)(s+3-s), \\ (s-1)^2(s+3)(-6-(s-2)s) \end{array} \right\}$$

Slide 7

Example 9.3 Continued II

$$= \gcd \left\{ \begin{array}{l} (s+3)(s-1)^2(s)(s+1), \\ (s+3)^2(s-1)(3), \\ (s-1)^2(s+3)(-1)(s^2-2s+6) \end{array} \right\}$$
$$= (s+3)(s-1)$$

$$\Rightarrow \Lambda(s) = \begin{bmatrix} 1 & 0 \\ 0 & (s-1)(s+3) \\ 0 & 0 \end{bmatrix}$$

SMITH-McMILLAN FORM

- Let $P(s)$ be a matrix of rational transfer functions. Let $P(s) = [P_{ij}(s)]$ have rank r . Then $P(s)$ can be transformed by elementary row and column operations to a pseudo diagonal rational matrix i.e. there exist unimodular $U(s), V(s)$ with

$$P(s) = U(s)M(s)V(s)$$

$$M(s) = \left[\begin{array}{cc|c} \frac{\epsilon_1(s)}{\psi_1(s)} & 0 & 0 \\ & \cdot & \cdot \\ & \frac{\epsilon_r(s)}{\psi_r(s)} & \\ & 0 & 0 \end{array} \right] \begin{array}{l} r \\ m-r \end{array}$$

$$\begin{array}{c|c} r & p-r \end{array}$$

Also $\epsilon_i(s) \mid \epsilon_{i+1}(s) \quad i = 1, \dots, r-1$

$\psi_{i+1}(s) \mid \psi_i(s) \quad i = 1, \dots, r-1$

$\epsilon_i(s), \psi_i(s)$ are monic

$\epsilon_i(s), \psi_i(s)$ are coprime $i, i = 1, \dots, r$

Slide 9

SMITH-McMILLAN FORM Continued

$M(s)$ is called the Smith-McMillan form (SMF) of $P(s)$ and is unique ; $U(s), V(s)$ are not unique.

- Even if $P(s)$ is proper, $M(s)$ need not be. Often there is a value $r_* < r$ (called the subrank) with $\psi_k(s) = 1, k \geq r_*$

Proof of SMF

Let $d(s)$ = least common multiple of the denominators in $P(s)$. So write $P(s) = \frac{N(s)}{d(s)}$

The Smith form of $N(s)$ gives

$$P(s) = U(s) \left[\begin{array}{ccc|c} \frac{\lambda_1(s)}{d(s)} & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \frac{\lambda_r(s)}{d(s)} & 0 \\ \hline \cdots & \cdots & 0 & 0 \end{array} \right] V(s)$$

Now cancel out common factors in $\lambda_i(s)/d(s)$

$i = 1, \dots, r$ to get $\frac{\lambda_i(s)}{d(s)} = \frac{\epsilon_i(s)}{\psi_i(s)}$ $i = 1, \dots, r$. Since $\lambda_i(s) \mid \lambda_{i+1}(s)$ the division properties follow.

SMF Example 9.4

$$P(s) = \begin{bmatrix} \frac{1}{s^2+3s+2} & \frac{-1}{s^2+3s+2} \\ \frac{s^2+s-4}{s^2+3s+2} & \frac{2s^2-s-8}{s^2+3s+2} \\ \frac{s-2}{s+1} & \frac{2s-4}{s+1} \end{bmatrix}$$

$$\Rightarrow d(s) = (s+1)(s+2)$$

$$\Rightarrow N(s) = \begin{bmatrix} 1 & -1 \\ s^2 + s - 4 & 2s^2 - s - 8 \\ s^2 - 4 & 2s^2 - 8 \end{bmatrix}$$

Example 9.4 Continued

Find Smith Form

$$\Delta_0(s) = 1$$

$$\Delta_1(s) = \gcd(\text{all elements}) = 1$$

$$\Delta_2(s) = \gcd(\text{all } 2 \times 2 \text{ minors})$$

$$= \gcd \left\{ \begin{vmatrix} 1 & -1 \\ s^2 + s - 4 & 2s^2 - s - 8 \end{vmatrix}, \right.$$

$$\begin{vmatrix} 1 & -1 \\ s^2 - 4 & 2s^2 - 8 \end{vmatrix},$$

$$\left. \begin{vmatrix} s^2 + s - 4 & 2s^2 - s - 8 \\ s^2 - 4 & 2s^2 - 8 \end{vmatrix} \right\}$$

$$= \gcd(3s^2 - 12, 3s^2 - 12, 8s^2 - 32)$$

$$= s^2 - 4$$

Example 9.4 Continued II

$$\Rightarrow \Lambda(s) = \begin{bmatrix} 1 & 0 \\ 0 & s^2 - 4 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \frac{\Lambda(s)}{d(s)} = \begin{bmatrix} \frac{1}{(s+1)(s+2)} & 0 \\ 0 & \frac{s^2-4}{(s+1)(s+2)} \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{(s+1)(s+2)} & 0 \\ 0 & \frac{s-2}{s+1} \\ 0 & 0 \end{bmatrix}$$

Smith Form Test for Coprimeness

$N(s), D(s)$ are right coprime if the Smith form for

$$\begin{bmatrix} D(s) \\ N(s) \end{bmatrix} \text{ is } \begin{bmatrix} I \\ 0 \end{bmatrix}$$

Proof. By the *gcrd* algorithm there is a unimodular $U(s)$ with

$$U(s) \begin{bmatrix} D(s) \\ N(s) \end{bmatrix} = \begin{bmatrix} R(s) \\ 0 \end{bmatrix} = \begin{bmatrix} I \\ 0 \end{bmatrix} R(s)$$

If $D(s), N(s)$ are right coprime any *gcrd* must be unimodular $\Rightarrow R(s)$ is unimodular $\Rightarrow R^{-1}(s)$ is a polynomial matrix so,

$$U(s) \begin{bmatrix} D(s) \\ N(s) \end{bmatrix} R^{-1}(s) = \begin{bmatrix} I \\ 0 \end{bmatrix} = \text{Smith Form}$$

Kalman SS form

- Using the *SMF* we can generalise Gilbert's method to allow $d(s)$ to have repeated roots. We suppose $P(s)$ is strictly proper.

Start with the SMF: $P(s) = U(s)M(s)V(s)$.

Represent $U(s)$ by columns and $V(s)$ by rows.

$$\begin{aligned}
 \Rightarrow P(s) &= [u_1(s), \dots, u_p(s)] \times \\
 &\left[\begin{array}{ccc|c} \frac{\epsilon_1(s)}{\psi_1(s)} & \dots & \dots & \\ \vdots & \vdots & \vdots & \\ & & \frac{\epsilon_r(s)}{\psi_r(s)} & 0 \\ & & 0 & 0 \end{array} \right] \left[\begin{array}{c} v_1^T(s) \\ \vdots \\ v_m^T(s) \end{array} \right] \\
 &= \sum_1^r u_k(s) v_k^T(s) \frac{\epsilon_k(s)}{\psi_k(s)} \\
 \Rightarrow P_{ij}(s) &= \sum_1^r u_k^{(i)}(s) v_k^{(j)}(s) \frac{\epsilon_k(s)}{\psi_k(s)} \\
 \Rightarrow P_{ij}(s) &= \sum_1^r \frac{b_k^{(ij)}(s)}{\psi_k(s)}, \text{ say}
 \end{aligned}$$

- Denote $n_k = \text{degree}(\psi_k(s))$

Slide 15

Kalman SS form Continued

Now $\frac{b_k^{(ij)}(s)}{\psi_k(s)}$ may not be strictly proper so divide out $\psi_k(s)^*$ to get

$$\frac{b_k^{(ij)}(s)}{\psi_k(s)} = \frac{\bar{b}_k^{(ij)}(s)}{\psi_k(s)} + PM_k^{(ij)}(s)$$

where the first term is strictly proper and the second is a polynomial matrix

$$\Rightarrow P_{ij}(s) = \sum_1^r \frac{\bar{b}_k^{(ij)}(s)}{\psi_k(s)} + \sum_1^r PM_k^{(ij)}(s)$$

However since $P_{ij}(s)$ is strictly proper we must have

$$\sum_1^r PM_k^{(ij)}(s) = 0$$

So only the strictly proper part matters and from it we can get the minimal SS form.

*We carry out the division in several stages.

Kalman SS form Continued II

We use three computations on SMF:

(i) By polynomial division we can write

$$\begin{aligned}\frac{u_k^{(i)}(s)}{\psi_k(s)} &= \frac{\bar{u}_k^{(i)}(s)}{\psi_k(s)} + \Pi_k^{(i)}(s) \\ &= \frac{u_{k,i}^T p_{-}(s)}{\psi_k(s)} + \Pi_k^{(i)}(s) \text{ see appendix} \\ \Rightarrow \frac{u_k(s)}{\psi_k(s)} &= \frac{C_k^T p_{-k}(s)}{\psi_k(s)} + \Pi_k(s)\end{aligned}$$

Note then that C_k is easily found. C_k^T is a $p \times n_k$ matrix; $p_{-k}(s)$ is an $n_k \times 1$ vector

(ii) Similarly we find

$$\frac{v_k^T(s) \epsilon_k(s)}{\psi_k(s)} = \frac{q_{+k}^T(s) B_k}{\psi_k(s)} + \Pi_k(s)$$

where again B_k is easily found. B_k is an $n_k \times m$ matrix; $q_{+k}(s)$ is an $n_k \times 1$ vector.

Kalman SS form Continued III

(iii) Let A_k be the $n_k \times n_k$ top companion form (or controller form) matrix corresponding to $\psi_k(s)$. Then from the Appendix,

$$(sI - A_k)^{-1} = \frac{p_{-k}(s)q_{+k}^T(s)}{\psi_k(s)} - U_k(s)$$

We find successively

$$\begin{aligned} P(s) &= \sum_1^r \frac{u_k(s)}{\psi_k(s)} v_k^T(s) \epsilon_k(s) \\ &= \sum_1^r \left(\frac{C_k^T p_{-k}(s)}{\psi_k(s)} + \Pi_k(s) \right) \epsilon_k(s) v_k^T(s) \\ &= \sum_1^r \left[C_k^T p_{-k}(s) \frac{v_k^T(s) \epsilon_k(s)}{\psi_k(s)} + M_k(s) \right] \end{aligned}$$

where $M_k(s)$ denotes a generic polynomial matrix

Kalman SS form Continued IV

Continuing

$$\begin{aligned}
 &= \sum_1^r \left\{ C_k^T p_{-,k}(s) \left[\frac{q_{+,k}^T(s) B_k}{\psi_k(s)} + \bar{\Pi}_k(s) \right] + M_k(s) \right\} \\
 &= \sum_1^r \left[C_k^T \frac{p_{-,k}(s) q_{+,k}^T(s)}{\psi_k(s)} B_k + M_k(s) \right] \\
 &= \sum_1^r \{ C_k^T [(sI - A_k)^{-1} + U_k(s)] B_k + M_k(s) \} \\
 &= \sum_1^r \{ C_k^T (sI - A_k)^{-1} B_k + M_k(s) \}
 \end{aligned}$$

Since again the first term is strictly proper we deduce $\sum_1^r M_k(s) = 0$ and so

$$P(s) = \sum_1^r C_k^T (sI - A_k)^{-1} B_k$$

Kalman SS form Continued V

And this yields the SS representation

$$\begin{aligned} A &= \text{blockdiag}(A_1, \dots, A_r) \\ B &= \begin{bmatrix} B_1 \\ \vdots \\ B_r \end{bmatrix}, C^T = [C_1^T, \dots, C_r^T] \end{aligned}$$

The SS dimension is

$$\begin{aligned} n &= \sum_{k=1}^r n_k \\ &= \sum_{k=1}^r \deg(\psi_k(s)) \\ &= \text{McMillan degree} \end{aligned}$$

Note that

$$a(s) = |sI - A| = \Pi_1^r |sI - A_k| = \Pi_1^r \psi_k(s)$$

Kalman SS form - Example 9.5

$$P(s) = \begin{pmatrix} s^2 + 6 & s^2 + s + 4 \\ 2s^2 - 7s - 2 & s^2 - 5s - 2 \end{pmatrix} \frac{1}{(s-1)(s+1)(s+2)}$$

Note $P(s)$ is strictly proper. First find SMF

$$[d(s) = (s-1)(s+1)(s+2) = s^3 + 2s^2 - s - 2]$$

$$\Delta_0(s) = 1$$

$$\Delta_1(s) = \gcd(\text{all elements})=1$$

$$\begin{aligned} \Delta_2(s) &= \begin{vmatrix} s^2 + 6 & s^2 + s + 4 \\ 2s^2 - 7s - 2 & s^2 - 5s - 2 \end{vmatrix} \\ &= (s^2 + 6)(s^2 - 5s - 2) - (2s^2 - 7s - 2)(s^2 + s + 4) \\ &= s^4 - 5s^3 - 2s^2 + 6s^2 - 30s - 12 \\ &\quad - [2s^4 + 2s^3 + 8s^2 - 7s^3 - 7s^2 - 28s - 2s^2 - 2s - 8] \\ &= -s^4 + 0.s^3 + 5s^2 + 0.s - 4 \\ &= -(s^4 - 5s^2 + 4) \\ &= -(s^2 - 4)(s^2 - 1) \end{aligned}$$

Slide 21

Example 9.5 Continued

$$\Rightarrow \lambda_1(s) = 1$$

$$\lambda_2(s) = -(s-2)(s+2)(s-1)(s+1)$$

$$\Rightarrow \Lambda(s) = \begin{pmatrix} 1 & 0 \\ 0 & -(s-1)(s+1)(s-2)(s+2) \end{pmatrix}$$

$$\Rightarrow M(s) = \begin{bmatrix} \frac{1}{(s-1)(s+1)(s+2)} & 0 \\ 0 & -(s-2) \end{bmatrix}$$

$$\frac{\epsilon_1(s)}{\psi_1(s)} = \frac{1}{(s-1)(s+1)(s+2)} = \frac{1}{s^3+2s^2-s-2}$$

$$\Rightarrow n_1 = \deg(\psi_1) = 3$$

$$\Rightarrow a_1, a_2, a_3 = 2, -1, -2$$

$$p_{-1}(s) = \begin{pmatrix} s^2 \\ s \\ 1 \end{pmatrix}, p_{+,1}(s) = \begin{pmatrix} 1 \\ s \\ s^2 \end{pmatrix}$$

$$q_{+,1}(s) = \begin{pmatrix} 1 \\ 2+s \\ -1+2s+s^2 \end{pmatrix}$$

Slide 22

Example 9.5 Continued II

$$\mathcal{A}_- = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{pmatrix}$$

$$\Rightarrow \mathcal{A}_-^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & -2 & 1 \end{pmatrix}$$

$$\frac{\epsilon_2(s)}{\psi_2(s)} = -(s - 2)$$

$$\Rightarrow n_2 = \deg(\psi_2) = 0!$$

Example 9.5 Continued III

Unimodular matrices are

$$\begin{aligned}U(s) &= (u_1, u_2) = \begin{bmatrix} 1 & 0 \\ \frac{s^3+2s^2-11s-2}{10} & 1 \end{bmatrix} \\V(s) &= \begin{pmatrix} v_1^T \\ v_2^T \end{pmatrix} = \begin{bmatrix} s^2+6 & s^2+s+4 \\ \frac{s+2}{10} & \frac{s+3}{10} \end{bmatrix} \\\frac{u_1(s)}{\psi_1(s)} &= \frac{1}{s^3+2s^2-s-2} \begin{pmatrix} 1 \\ \frac{s^3+2s^2-s-2-10s}{10} \end{pmatrix} \\&= \frac{1}{\psi_1(s)} \begin{pmatrix} 1 \\ -s \end{pmatrix} + PM(s) \\&= \frac{1}{\psi_1(s)} \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} s^2 \\ s \\ 1 \end{pmatrix} + PM(s) \\\Rightarrow C_1^T &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}\end{aligned}$$

Slide 24

Example 9.5 Continued IV

$$\begin{aligned}\frac{v_1^T(s)\epsilon_1(s)}{\psi_1(s)} &= (s^2 + 6, s^2 + s + 4) \frac{(+1)}{s^3 + 2s^2 - s - 2} \\&= \left(\frac{1, s, s^2}{\psi_1(s)} \right) \begin{pmatrix} 6 & 4 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \\&= \frac{q_{+,1}^T(s)}{\psi_1(s)} B_1 \\B_1 &= \mathcal{A}_-^{-T} \begin{pmatrix} 6 & 4 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \\&= \begin{pmatrix} 1 & -2 & 5 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 6 & 4 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \\&= \begin{pmatrix} 11 & 7 \\ -2 & -1 \\ 1 & 1 \end{pmatrix}\end{aligned}$$

Slide 25

Example 9.5 Continued V

So SS form is

$$A = A_1 = \begin{pmatrix} -2 & 1 & 2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$B = B_1 = \begin{pmatrix} 11 & 7 \\ -2 & -1 \\ 1 & 1 \end{pmatrix}$$

$$C^T = C_1^T = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

APPENDIX TO 9B: PROPERTIES OF COMPANION MATRIX

Introduce controller form companion matrix

$$\begin{aligned}
 A_c &= \begin{bmatrix} -a_1 & -a_2 & \cdots & \cdots & -a_n \\ 1 & 0 & \cdots & \cdots & 0 \\ 0 & 1 & \cdots & \cdots & \cdots \\ \vdots & \vdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix} \\
 p_-(s) &= \begin{bmatrix} s^{n-1} \\ s^{n-2} \\ \vdots \\ \vdots \\ 1 \end{bmatrix}, p_+(s) = \begin{bmatrix} 1 \\ s \\ \vdots \\ \vdots \\ s^{n-1} \end{bmatrix} \\
 q_+(s) &= \mathcal{A}_- p_+(s)
 \end{aligned}$$

APPENDIX Continued

e.g.

$$\begin{aligned}n &= 3 \\q_+(s) &= \begin{bmatrix} 1 & 0 & 0 \\ a_1 & 1 & 0 \\ a_2 & a_1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ s \\ s^2 \end{bmatrix} \\&= \begin{bmatrix} 1 \\ a_1 + s \\ a_2 + a_1 s + s^2 \end{bmatrix} \\a(s) &= s^n + a_1 s^{n-1} + \cdots + a_n\end{aligned}$$

APPENDIX Continued III: PROPERTIES

(i) $(sI - A_c)p_-(s) = e_1 a(s)$

(ii) $(sI - A_c)U(s) + I = e_1 q_+^T(s)$

$$U(s) = \begin{bmatrix} 0 & 1 & s & s^2 & \dots & \dots & s^{n-2} \\ 0 & 0 & 1 & s & \dots & \dots & s^{n-3} \\ 0 & 0 & 0 & 1 & \dots & \dots & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & s \\ 0 & \vdots & \vdots & \vdots & \vdots & \vdots & 1 \\ 0 & \vdots & \vdots & \vdots & \vdots & \vdots & 0 \end{bmatrix}$$

(iii) $(sI - A_c)^{-1} = \frac{p_-(s)q_+^T(s)}{a(s)} - U(s)$

(iv) If λ is a root of $a(s)$ then the corresponding right evec of A_c is $p_-(\lambda)$.

The left evec is $q_+(\lambda) = \mathcal{A}_- p_+(\lambda)$.

APPENDIX Continued IV

(v) For any polynomial

$$\begin{aligned}
 & \theta_1 s^{n-1} + \theta_2 s^{n-2} + \dots + \theta_n \\
 &= (\theta_1 \dots \theta_n) \begin{pmatrix} s^{n-1} \\ s^{n-2} \\ \vdots \\ 1 \end{pmatrix} \\
 &= \theta_+^T p_-(s) \\
 &= \theta_-^T p_+(s) \\
 &= \theta_-^T \mathcal{A}_-^{-1} \mathcal{A}_- p_+(s) \\
 &= \theta_*^T q_+(s), \theta_* = \mathcal{A}_-^{-T} \theta_-
 \end{aligned}$$

APPENDIX Continued V

Proofs: (v) is straight forward

Sketch proof for $n = 3$

(i)

$$\begin{aligned} A_c p_-(s) &= \begin{bmatrix} -a_1 & -a_2 & -a_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} s^2 \\ s \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -a_1 s^2 - a_2 s - a_3 \\ s^2 \\ s \end{bmatrix} \\ &= \begin{pmatrix} s^3 \\ s^2 \\ s \end{pmatrix} - (s^3 + a_1 s^2 + a_2 s + a_3) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\ &= sp_-(s) - a(s)e_1, \text{ as required} \end{aligned}$$

APPENDIX Continued VI

(ii)

$$\begin{aligned}
 A_c U(s) &= \begin{bmatrix} -a_1 & -a_2 & -a_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & s \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & -a_1 & -a_1 s - a_2 \\ 0 & 1 & s \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & s & s^2 \\ 0 & 1 & s \\ 0 & 0 & 1 \end{bmatrix} - e_1(1, a_1 + s, s^2 + a_1 s + a_2) \\
 &= I + sU(s) - e_1 q_+^T(s), \text{ as required}
 \end{aligned}$$

(iii) From (ii)

$$\begin{aligned}
 U(s) + (sI - A_c)^{-1} &= (sI - A_c)^{-1} e_1 q_+^T(s) \\
 &= \frac{p_-(s) q_+^T(s)}{a(s)}, \text{ by (1) as required}
 \end{aligned}$$

Slide 32

APPENDIX Continued VII

(iv) By (i) $(\lambda I - A_c)p_-(\lambda) = e_1 a(\lambda) = 0$ as required. Next

$$\begin{aligned} (iii) \Rightarrow \quad & a(s)I = p_-(s)q_+^T(s)(sI - A_c) \\ & - a(s)U(s)(sI - A_c) \\ \Rightarrow 0 &= p_-(\lambda)q_+^T(\lambda)(\lambda I - A_c) \\ \Rightarrow 0 &= q_+^T(\lambda)(\lambda I - A_c) \end{aligned}$$